

Tracing the Wedge

What am I attempting?

I am interested in creating a simulation which measures saltwater intrusion across bodies of water around New Orleans. I am curious about examining the impacts of saltwater (encroaching onto the Lake Pontchartrain Area Watershed from the Gulf of Mexico) on freshwater aquifers. How will ecologies shape as saltwater creeps further upstream? What industries will be impacted by the drastic salinization of freshwater waterscapes?

The Datasets

I will use USGS public water-monitoring locations across the Lower Mississippi. USGS Water Data APIs provide field measurements and location metadata output as CSV and GeoJSON. I would use temporal columns such as the date, something which measures salinization such as the discharge column, and the lat/long.

The Research

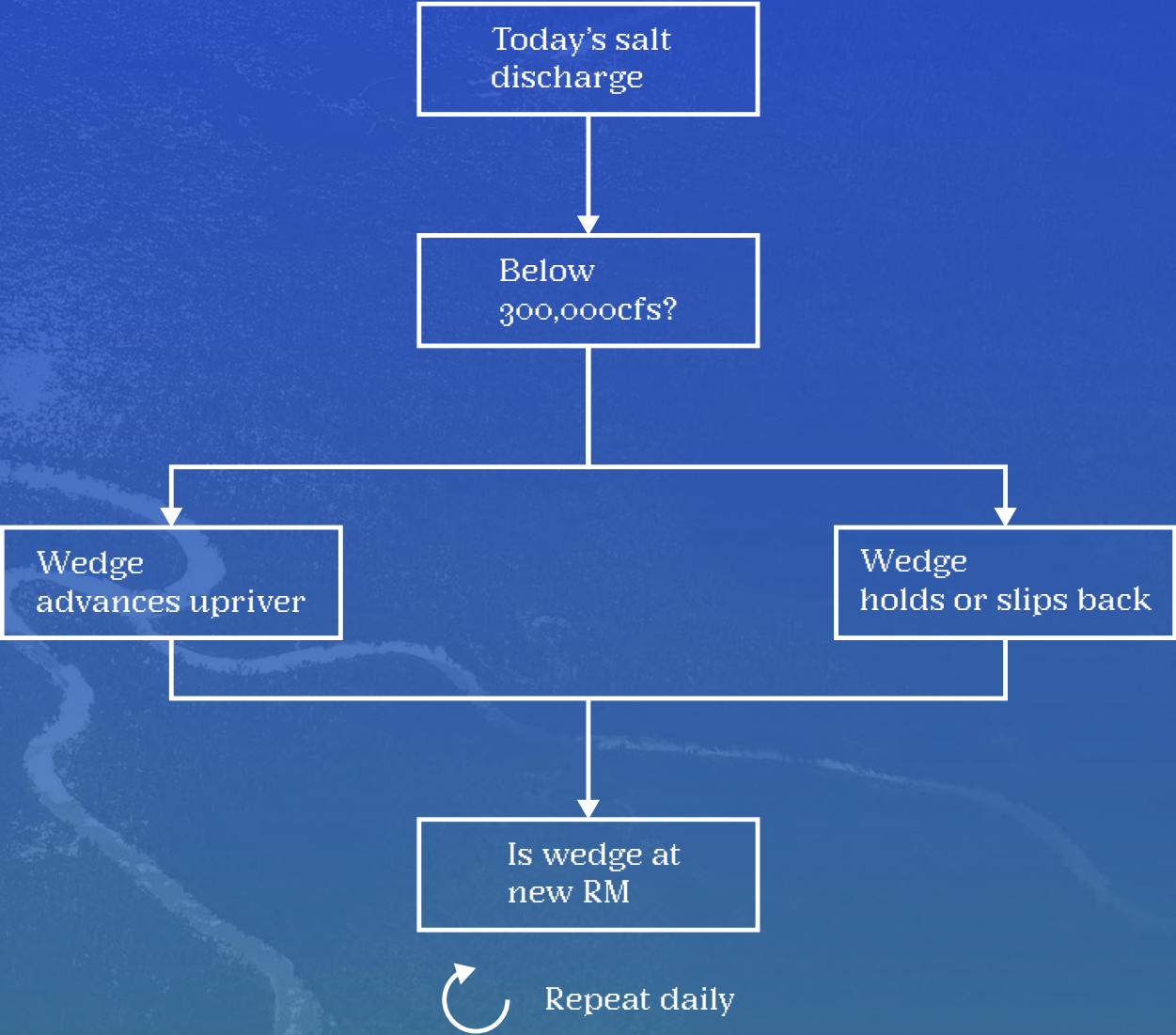
The most important source I have found is the USACE New Orleans District “Saltwater Wedge Now” tracker and impact timeline, which published the current toe position and forecast timeline. Other relevant sources include U.S. Army Corps of Engineers publications on the wedge and the Tulane School of Public Health paper “5 Things to Know about the Saltwater Intrusion of the Mississippi River”.

The Record

This simulation measures the position of the saltwater toe as it passes through the Mississippi River. Our data begins in August of 2023 - it is initially observed beginning at RM75 (River's Mouth marker). The rows are time-based, and the columns are the discharge of salinity and the location of the toe (tip of wedge).

The Rule

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The Run

This simulation will chronologically progress forward - salt discharge develops in conjunction with time, which is measured at each RM.

<i>date</i>	<i>discharge cfs</i>	<i>toe</i>	<i>surface front RM</i>	<i>miles to Belle Chasse</i>	<i>past 250mg ar gauge?</i>
2023-09-01	150,000	58	75	0	yes
2023-09-02	148,000	60.5	77.5	-2.5	yes
2023-09-03	152,000	62.5	79.5	-4.5	yes

Output

This data would generate a map of the river with two markers as points and the toe of the saltwater wedge represented almost like a slider - could be a vector/point cloud like thing or progression of raster imaging. As the timeline progresses, various items of data could be displayed as reaction to the move of the wedge.

Exemptions

There are many natural factors which could potentially impact this simulation, including tide, channel dredging depth, wind patterns, and most importantly, sea level rise (maybe this could be included into simulation).

August '23

Future

- XYZ Date:
- Discharge any%%
 - ToeRM
 - Miles to Belle Chasse